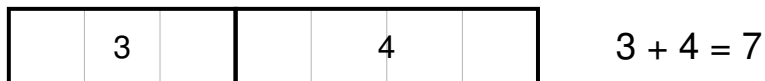
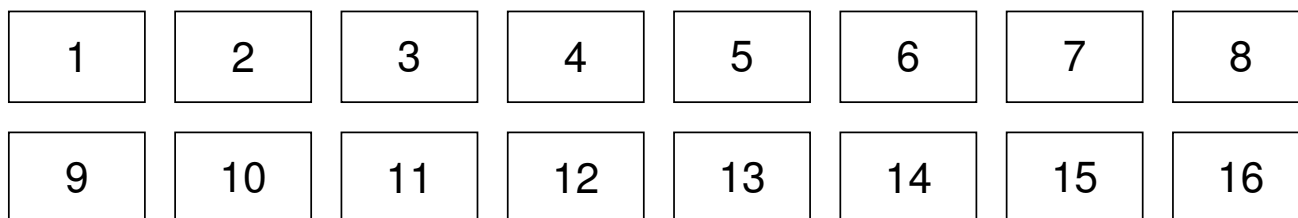


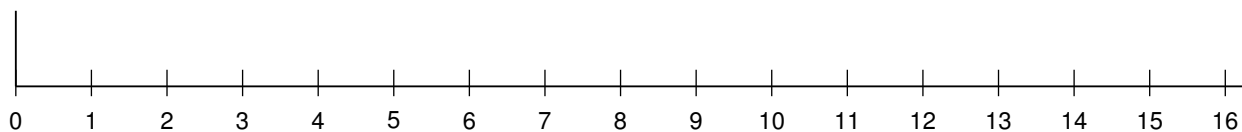
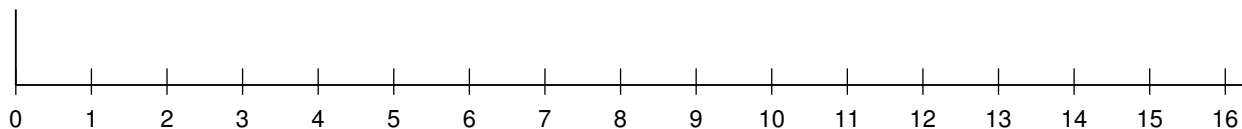
Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.



Problem 1: Use 3-rods and 4-rods. Find all the lengths from 1 through 16 that can be built, and mark the ones that cannot.



Mark a target on a ruler and build from 0 to the mark.



Problem 2: Find every way to make 12, 16 and 24 with 3-rods and 4-rods. Rod order does not count.

12

16

24

Problem 3: Use 3-rods and 5-rods. Find all the impossible lengths from 1 through 20. Is there a last impossible length?

	3	
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		5		
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1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	

Problem 4: For each pair of rods, build a small collection that convinces someone that every target after a certain length can be made. Say where the gaps stop.

	3	
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		4	
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	3	
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		5		
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Problem 5: Use 4-rods and 7-rods. Find the last impossible length, and explain why there are no later gaps.



12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27
28							

Problem 6: Someone says every pair of rod lengths eventually has no more gaps. Test 2 and 4, then 4 and 7. Is the claim true?